

Small Molecule Screens

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BINF5501 – Bioinformatics & Precision Medicine

Lecture 9



HUMBER

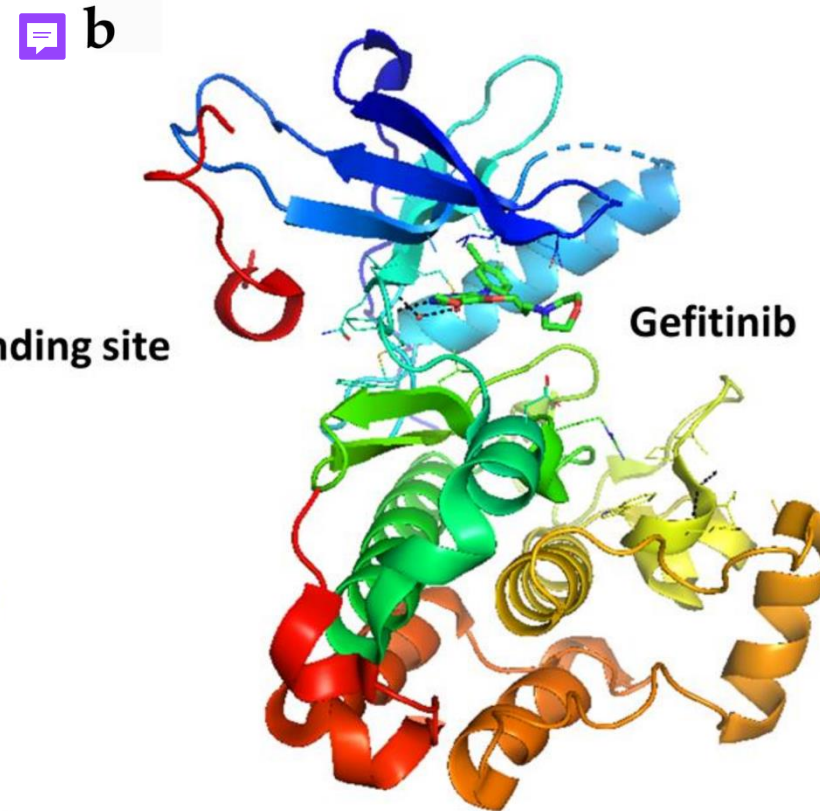
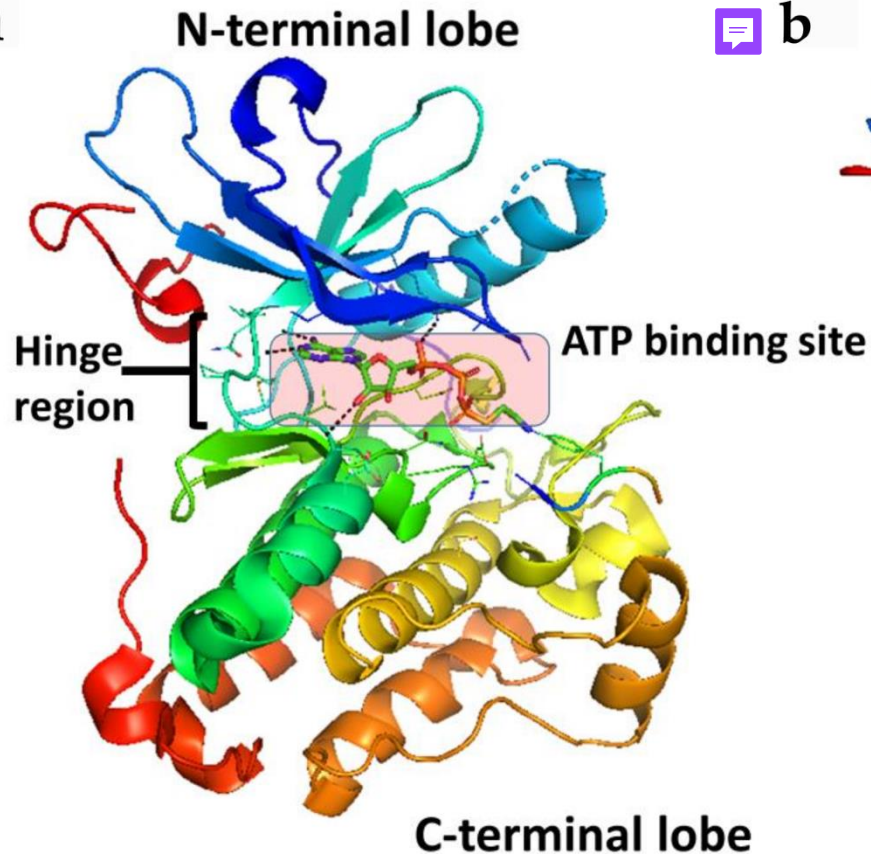
Faculty of Health Sciences & Wellness

Today's Lecture and Discussion Topics

- Small molecule drugs and their mechanisms of action
- Cancer cell line drug sensitivity screens
- Gene expression-based small molecule screens

Small molecule drugs often interact with protein targets and affect their function

 **a**
EGFR kinase domain



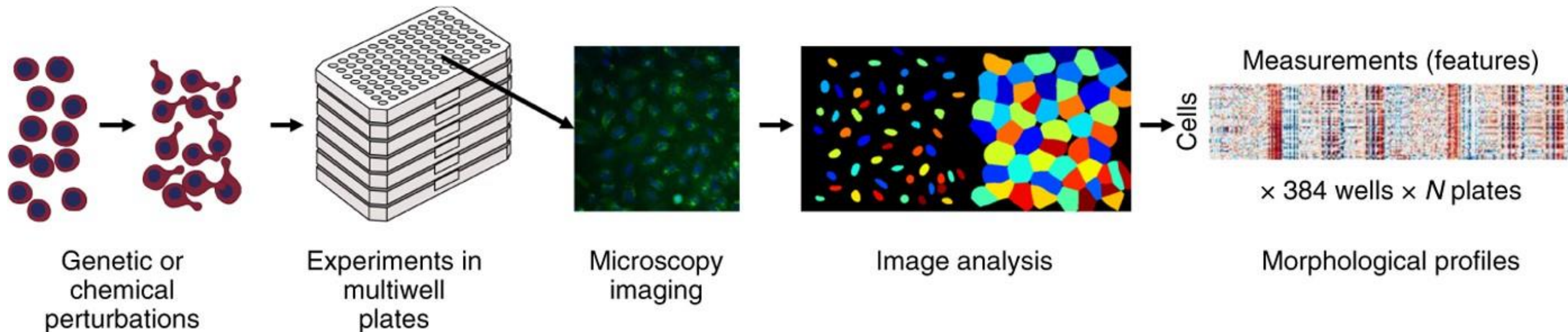
Mechanisms of action of small molecules

- **Enzyme Inhibitors:** These drugs block the activity of specific enzymes, thereby interfering with disease processes. 
- **Receptor Agonists:** Activate receptors to mimic the action of a natural ligand. 
- **Receptor Antagonists:** Block receptors to inhibit the action of a natural ligand.
- **Ion Channel Modulators:** These drugs regulate the flow of ions across cell membranes, which is crucial for processes like neurotransmission and muscle contraction. 
- **Protein-Protein Interaction Modulators:** These drugs disrupt or stabilize interactions between proteins, which can be critical in pathways like cell signaling and apoptosis
- **Allosteric Modulators:** These drugs bind to a site on a protein other than the active site, causing a conformational change that affects the protein's activity. 
- **Bifunctional Molecules (PROTACs):** These are designed to target proteins for degradation by linking a target protein to an E3 ubiquitin ligase, leading to the protein's destruction by the proteasome. 
- **Nucleic Acid Interacting Drugs:** These drugs interact with DNA or RNA to inhibit processes like transcription or translation. 

Phenotypic screening is target-agnostic



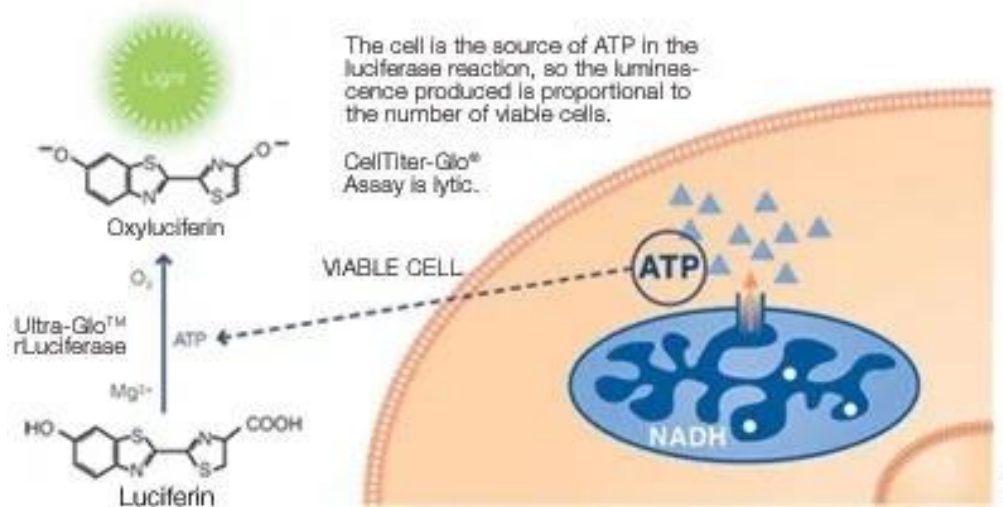
Cell Painting



High-throughput assays of cell viability

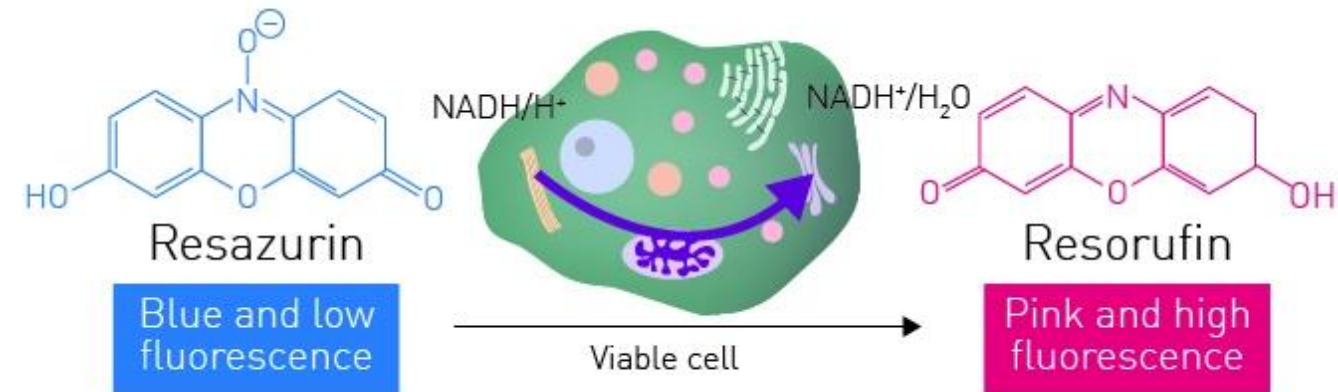


Cell Titer-Glo



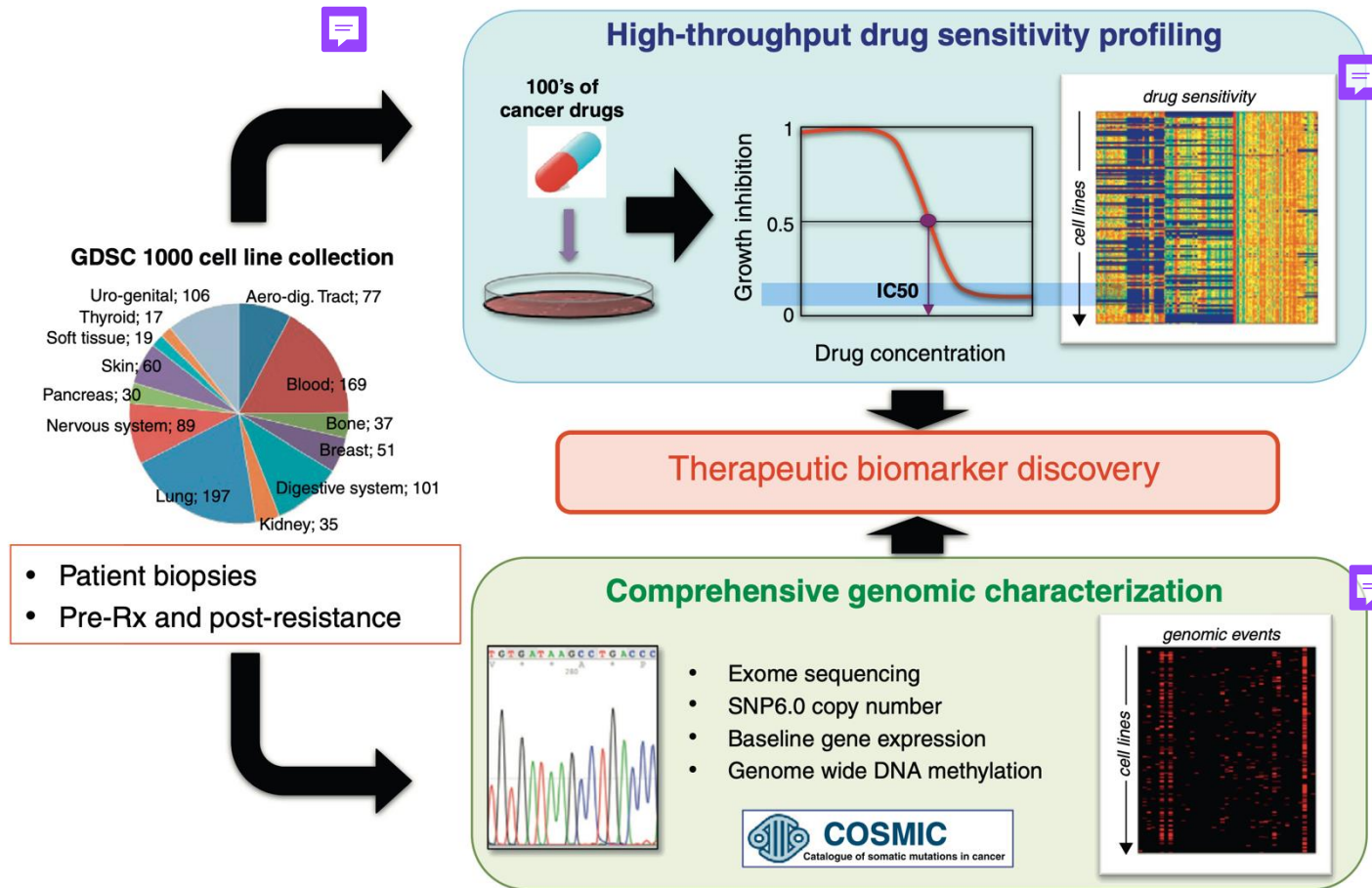
https://www.promega.ca/products/cell-health-assays/cell-viability-and-cytotoxicity-assays/celltiter_glo-luminescent-cell-viability-assay/?catNum=G7570

Alamar Blue

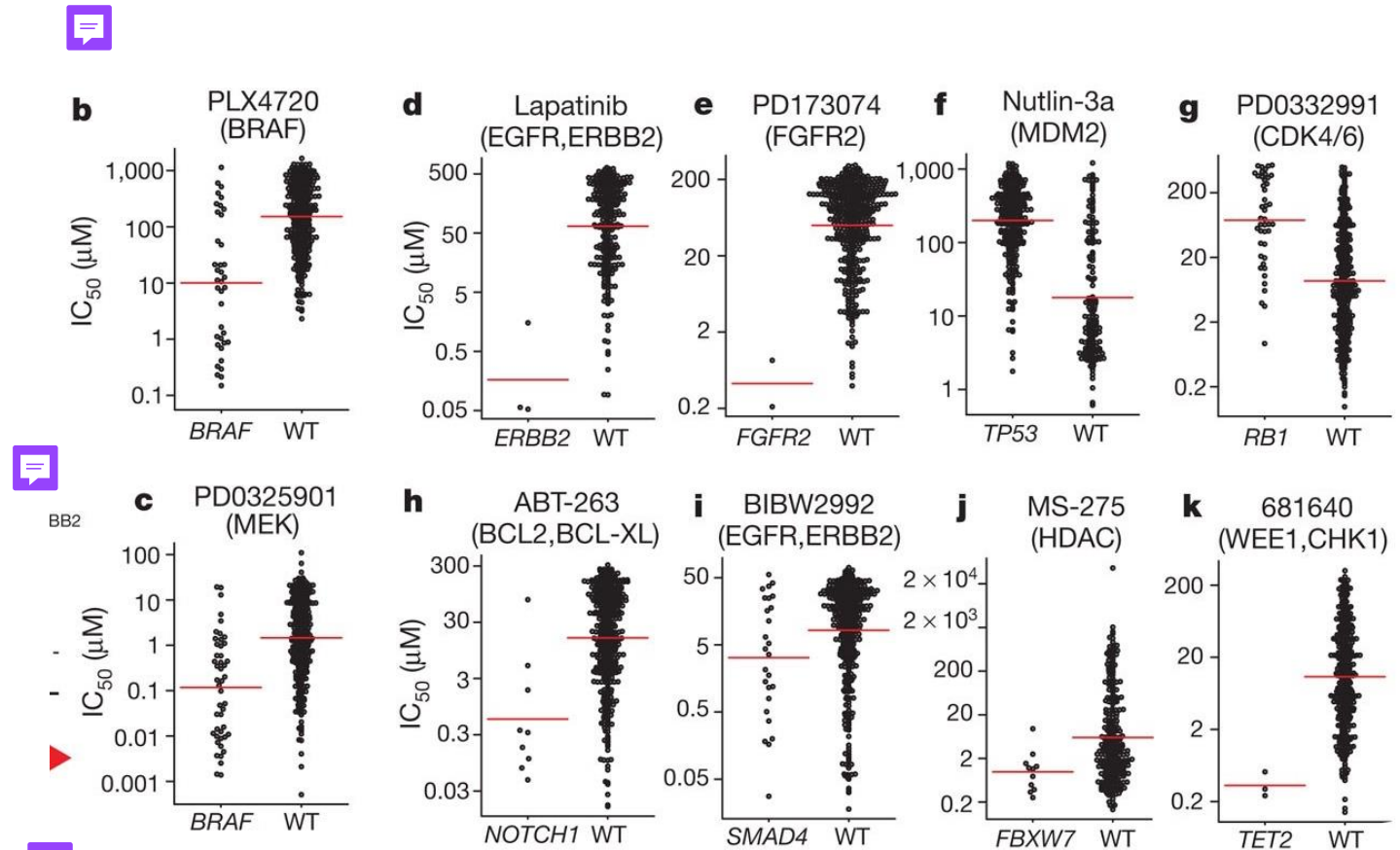
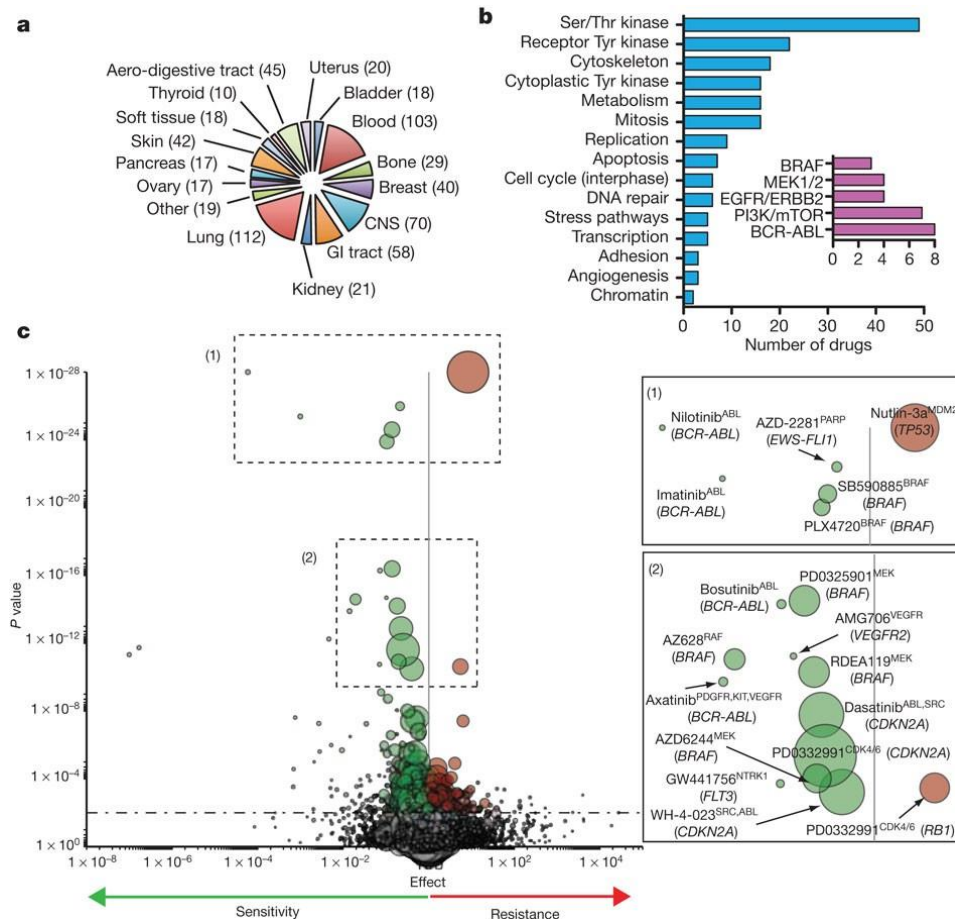


<https://www.bmglabtech.com/en/application-notes/resazurin-cell-viability-assay-for-the-discovery-of-selective-cytotoxic-compounds-in-canine-mammary-cancer-research/>

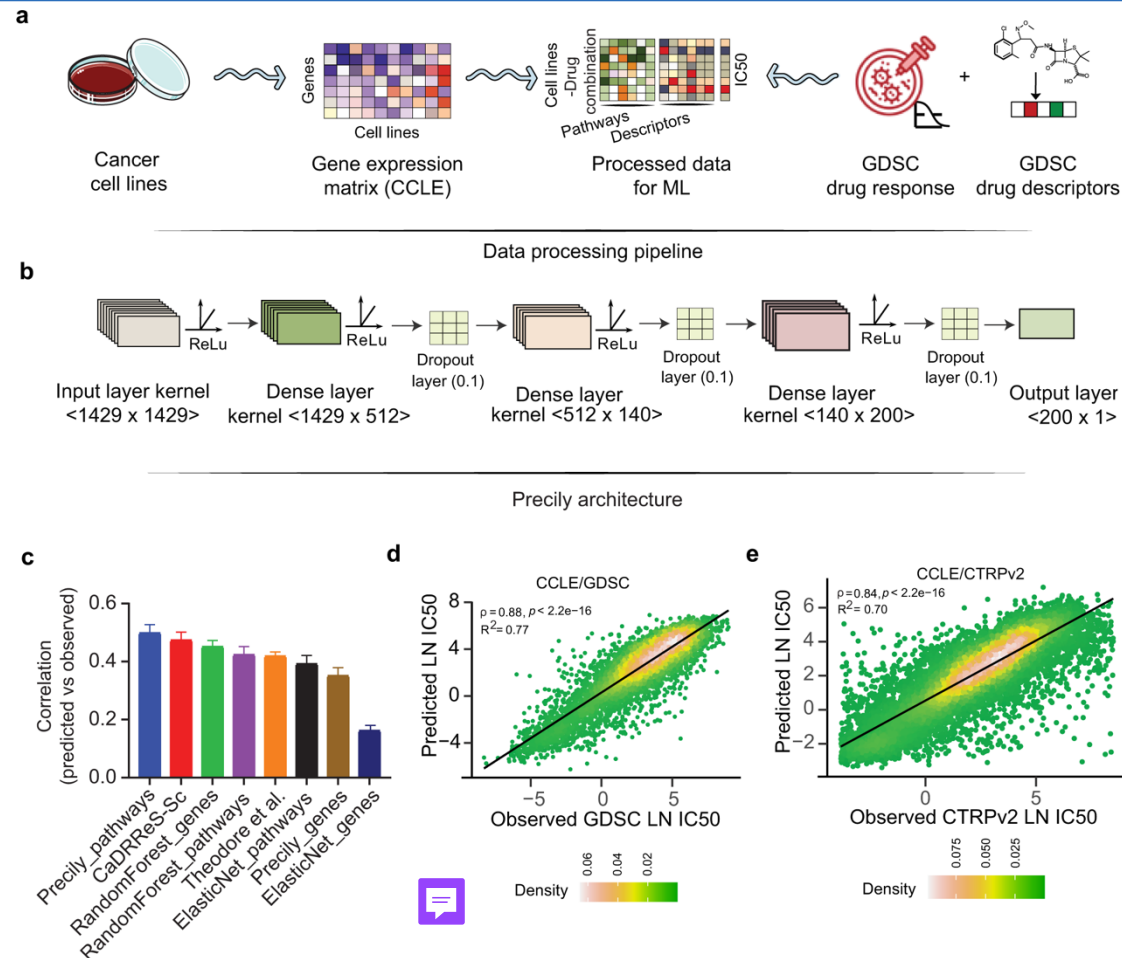
Systematic cancer cell line drug screens



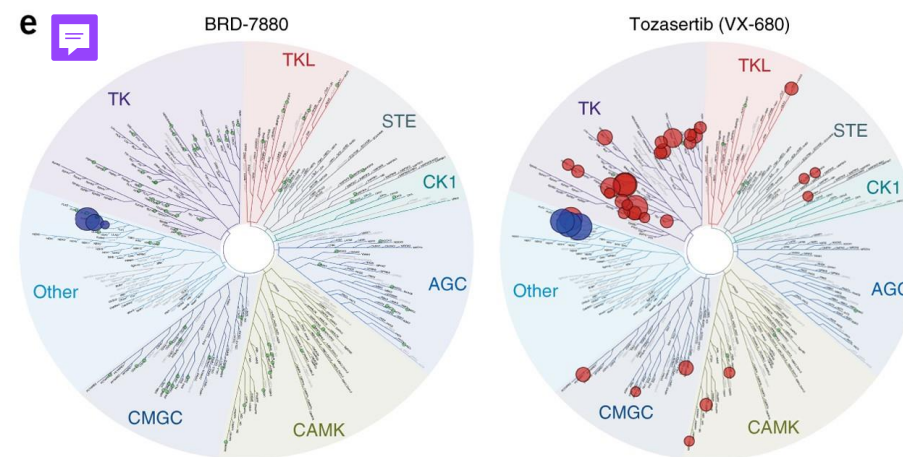
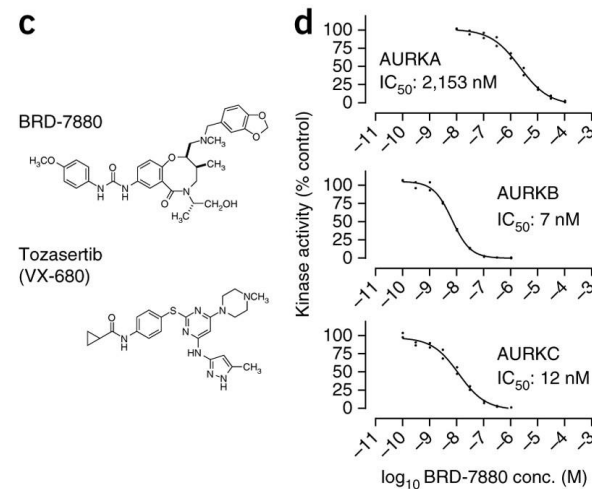
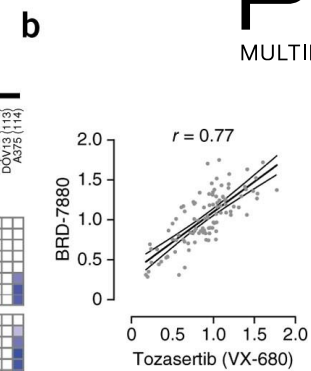
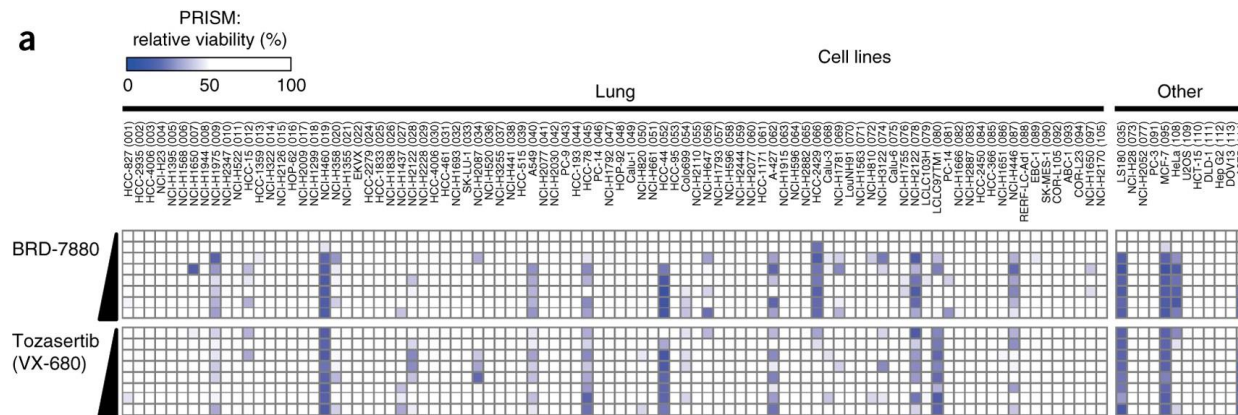
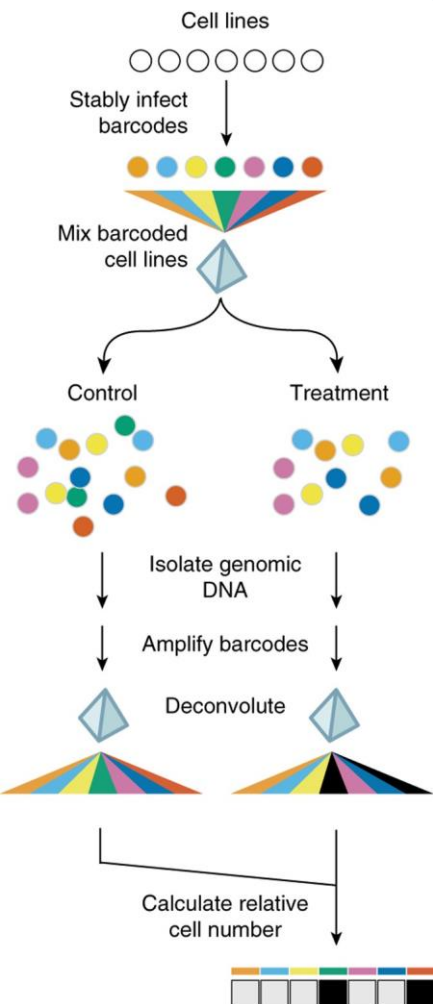
Mutations as biomarkers of drug sensitivity



Pre-existing gene expression patterns predict drug sensitivity



Pooled cell viability screens enable hundreds of barcoded cell lines to be assayed in parallel



DepMap enables systematic identification of biomarkers and molecular targets



depmap portal

 Search for a gene, cell line, compound, lineage, or file

Tools ▾

Downloads

About ▾

Forum

Custom Analyses

Select type of analysis to run

- ☒ **Pearson correlation**
Computes Pearson correlation for each feature in the selected dataset along with corresponding q-value.
- ☐ **Two class comparison**
Computes a moderated estimate of the difference between groups' means for each feature along with the corresponding q-value.

2. Select a dataset

- ☒ Portal data
- CRISPR (DepMap Public 24Q4+Score, Chronos) ▾
- ☐ Custom upload

1. Select a data slice:

- ☒ Portal data

Compound	▾
NUTLIN-3A (-)	▾
Drug sensitivity AUC (Sanger GDSC2)	▾
NUTLIN-3A (-) (GDSC2:1047)	× ▾

- ☐ Custom upload

3. Select cell lines to run on:

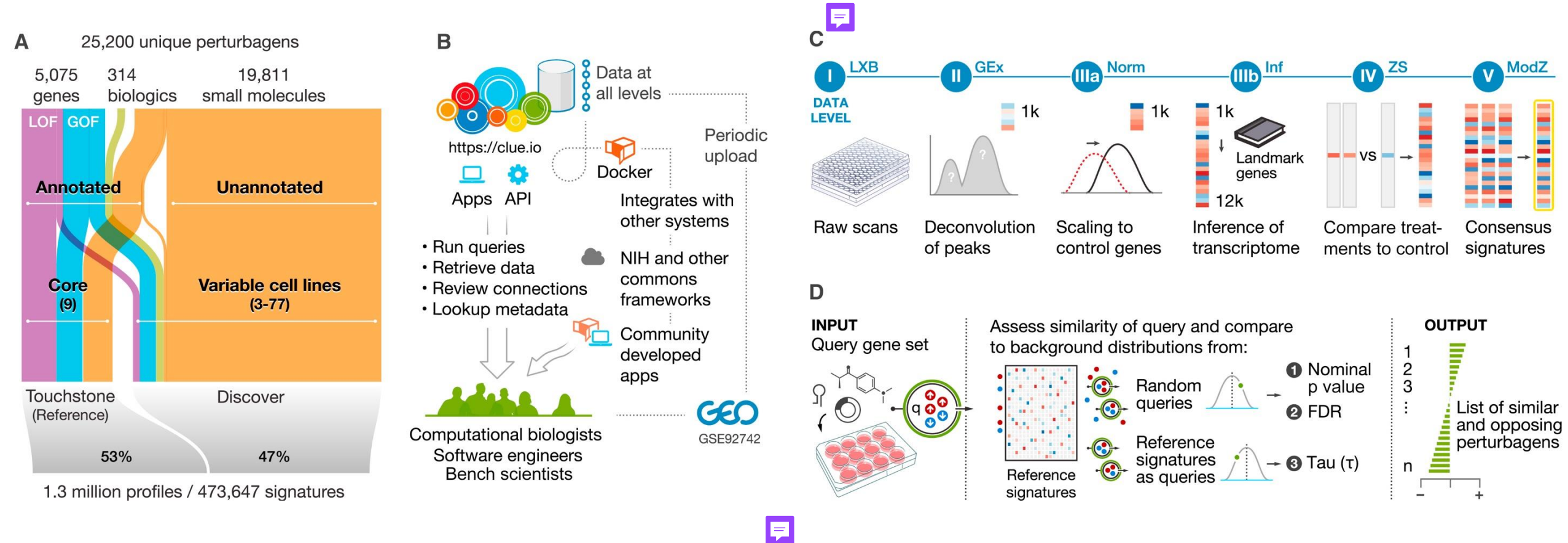
- ☒ Use all cell lines
- ☐ Select a subset of cell lines

Run

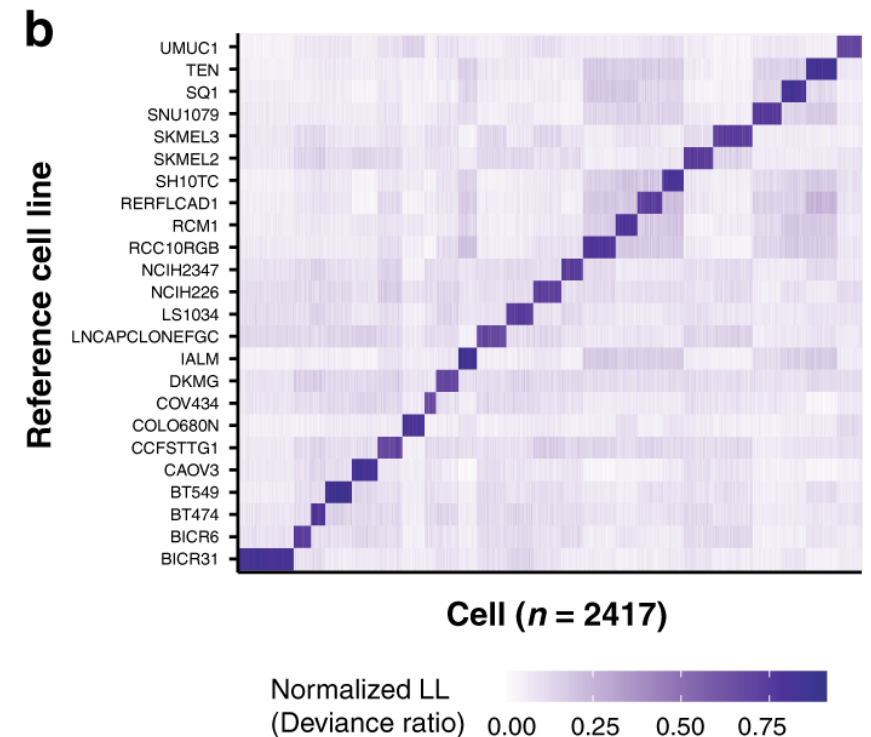
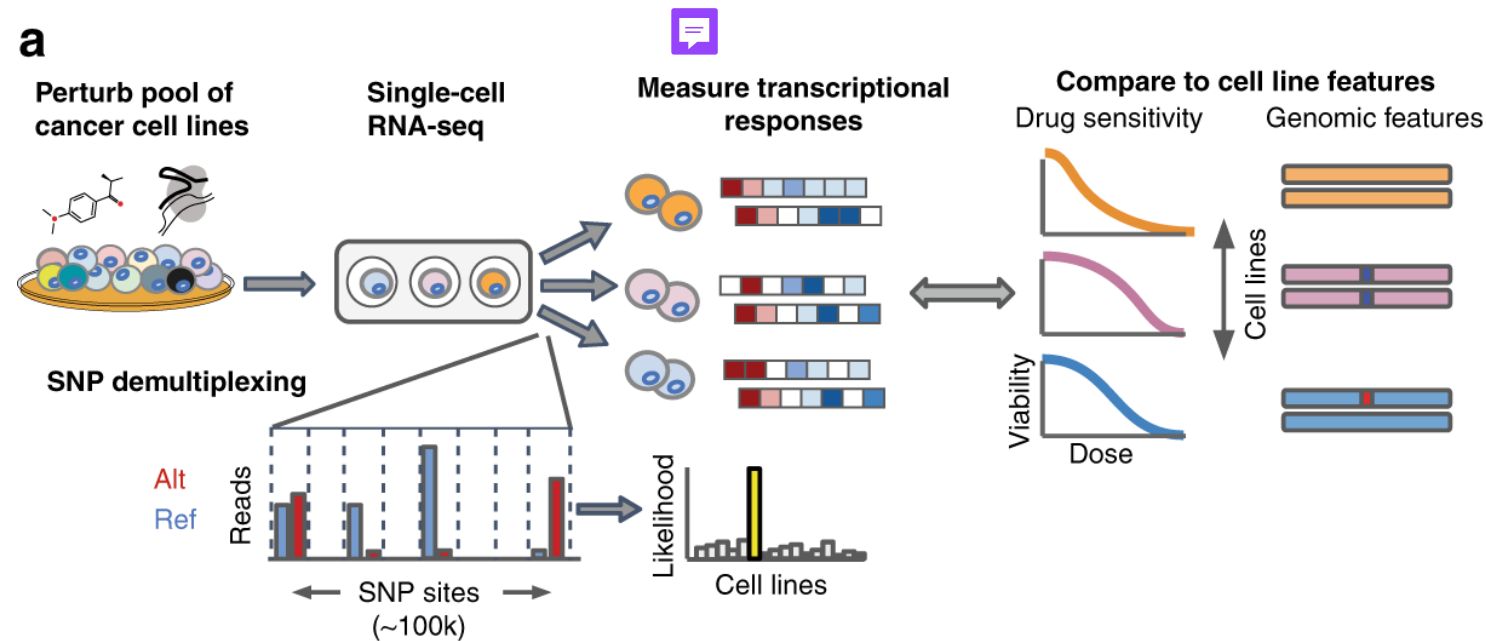
Drug mechanism of action can be predicted based on changes in gene expression

– L1000 and The Connectivity Map

Subramanian et al. Cell, Volume 171, Issue 6, 1437 - 1452.e17
<https://www.cell.com/cell/fulltext/S0092-8674%2817%2931309-0>



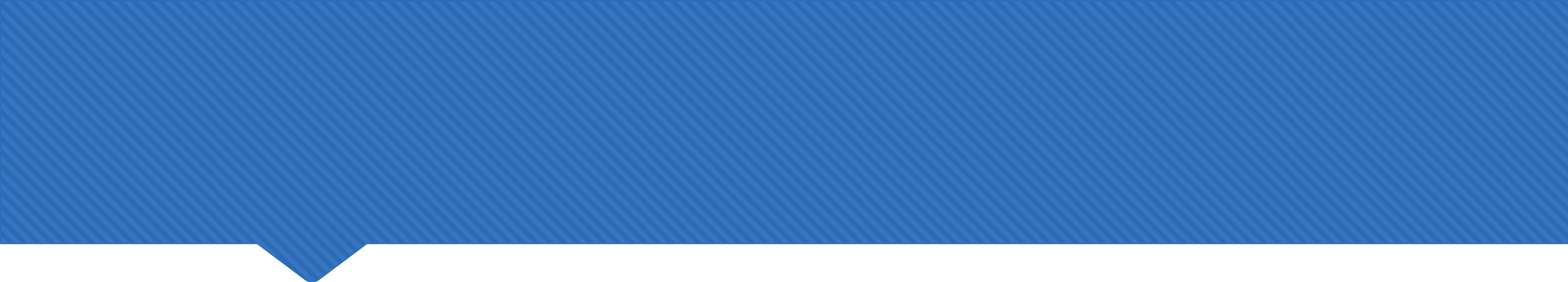
Multiplexed single-cell transcriptional response profiling to define cancer vulnerabilities and therapeutic mechanism of action





Summary

- Phenotypic screens can elucidate molecular targets and mechanisms of action of chemical and genetic perturbations
- High-throughput screens that measure drug-induced changes in cell viability have been performed in large collections of well-characterized human cancer cell lines to enable biomarker and molecular target discovery
- Drug sensitivity and mechanism of action can sometimes be inferred from pre-existing gene expression profiles or induced changes in gene expression



Questions?